Phoenix — Composite Blade Conversion System

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1. Executive Summary

Overview

Wind turbine blades are reaching end-of-life at accelerating scale, creating a growing disposal challenge with limited viable solutions. This is creating a rapidly growing disposal bottleneck across the energy sector.

Phoenix is a structured industrial system designed to convert turbine blade waste into usable material outputs using existing, proven processes and scalable deployment models.

Turbine blades are composed of fiberglass-reinforced polymer composites that cannot be remelted, are difficult to separate, and are increasingly restricted from landfill disposal. As disposal costs rise and infrastructure lags behind demand, a scalable solution is required.

Phoenix applies a staged process:

Cut → Reduce → Separate → Convert

This transforms composite blade waste into:

- Insulation materials
- Composite lumber
- Industrial filler

Business Model

Phoenix generates revenue through:

- **Tipping fees** (paid to accept blade waste)
- **Material sales** (converted outputs)

This creates a **dual-revenue system from a single input stream**.

Deployment

Phoenix is deployed through partnerships with:

- landfill operators
- waste management companies
- wind energy firms

The system operates within existing infrastructure, minimizing capital requirements.

Ask

Phoenix is seeking:

> A pilot partner with site access and turbine blade inventory to validate the system under real-world conditions.

Summary Statement

Phoenix converts turbine blade waste into usable materials through a scalable, partner-driven industrial system.

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2. Problem Definition

Overview

Wind energy deployment has scaled rapidly. As early-generation turbines reach end-of-life, a critical infrastructure challenge has emerged:

Industrial-scale disposal of turbine blades

Scale

- Tens of thousands of blades expected annually (U.S.)
- Millions of tons globally over the next decade
- Individual blades:
  - 100–200+ feet
  - 10–20 tons

Material Constraints

Wind turbine blades are composed of:

- Fiberglass reinforcement
- Thermoset resins
- Bonding materials

These materials:

- Do not melt or remold
- Do not biodegrade
- Are difficult to separate

Current Solutions

- **Landfilling** → constrained, costly
- **Cement co-processing** → low value
- **Experimental recycling** → not scalable

Core Problem

Blade disposal is:

- High volume
- Difficult to process
- Increasing in cost
- Growing over time

Existing solutions are **fragmented and insufficient at scale**.

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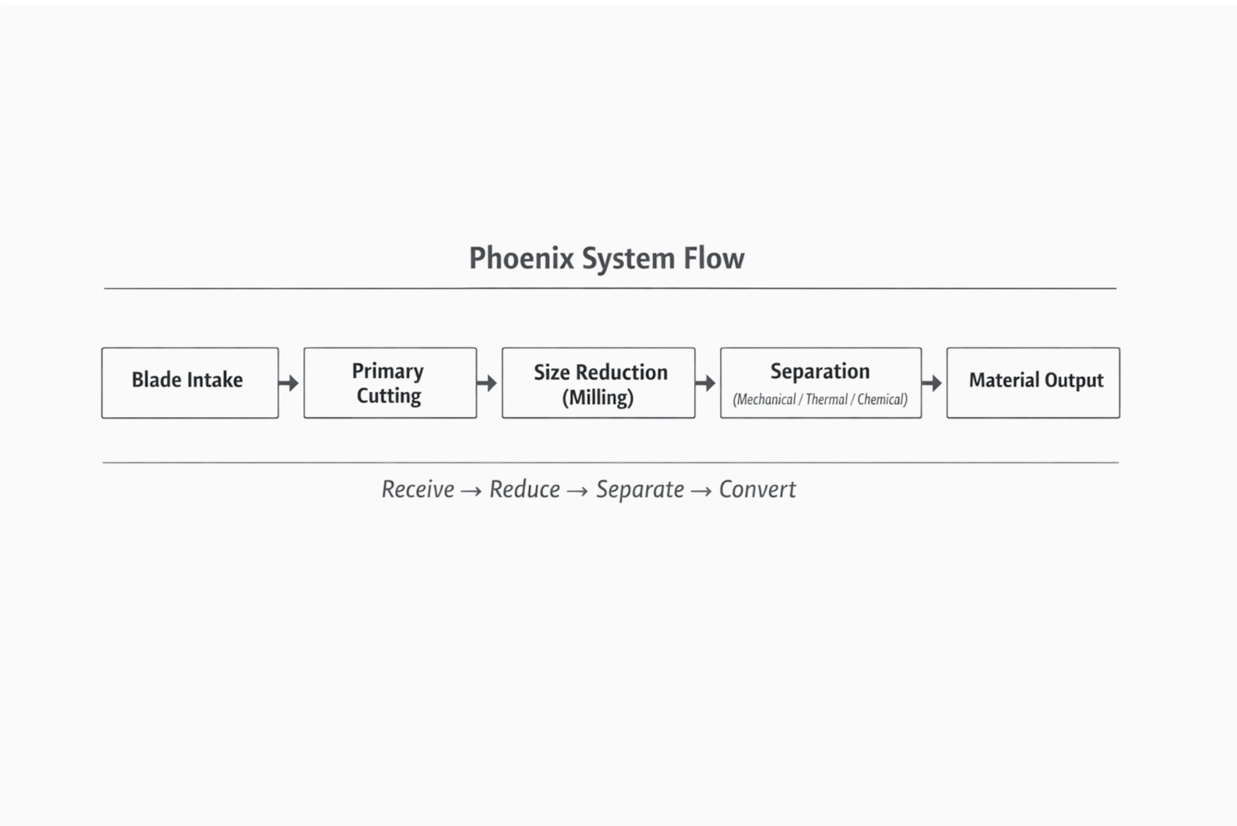
3. Process Overview

System Flow

Receive → Cut → Reduce → Separate → Output



Stages

- Intake and staging
- Primary cutting
- Secondary reduction

- Separation
- Material output

Design Principles

- Use existing equipment
- Modular process design
- Output-driven processing
- On-site deployment capability

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4. Separation Strategy

Methods

- **Mechanical** → low cost, high throughput
- **Thermal** → moderate cost, improved output
- **Chemical** → high quality, higher complexity

Summary

Method	Output Quality	Cost	Scalability
Mechanical	Low	Low	High
Thermal	Medium	Medium	High
Chemical	High	High	Medium

Key Insight

Phoenix selects the pathway based on cost-to-output ratio and target product specification.

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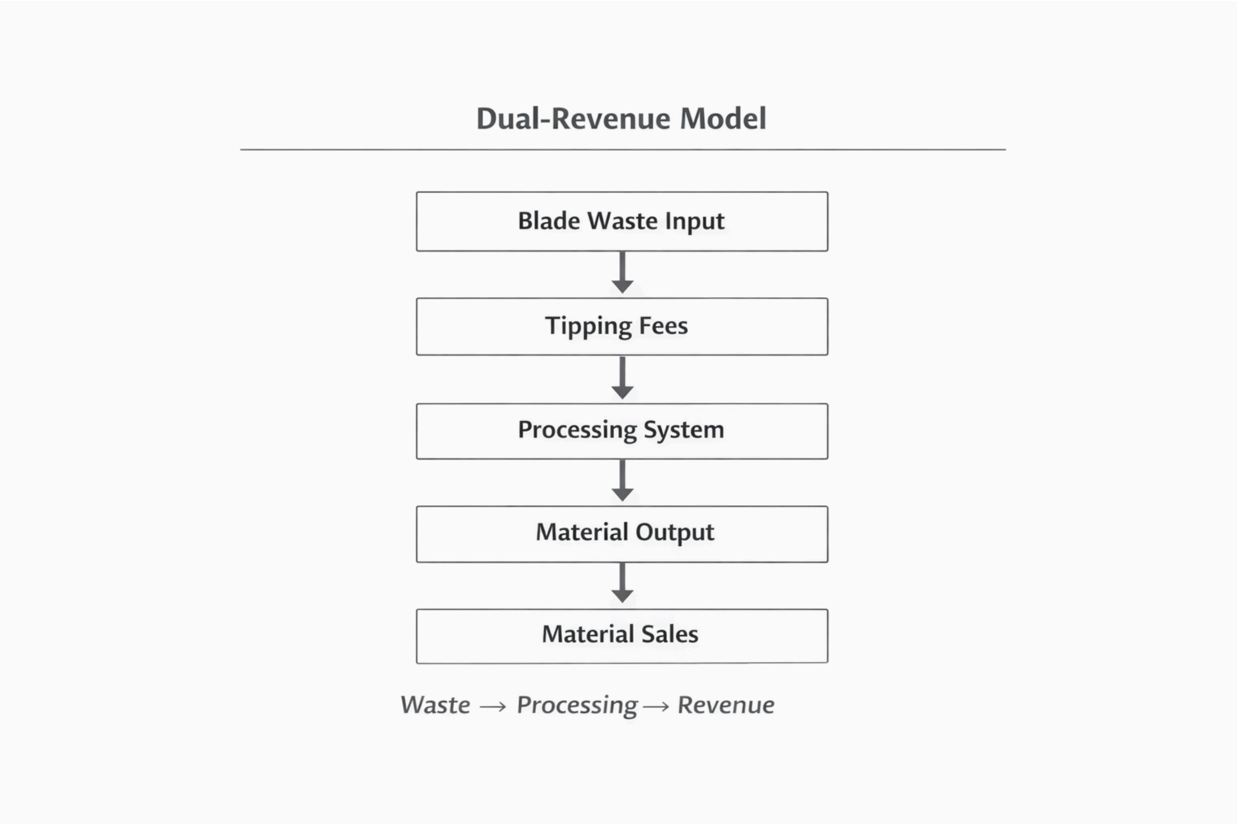
5. Product Outputs & Markets

Output Streams

- Insulation materials
- Composite lumber
- Industrial filler



Key Insight

Phoenix converts a single waste stream into multiple revenue-generating outputs.

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6. Business Model

Revenue Streams

- Tipping fees
- Material sales

Structure

Phoenix is licensed to operators.

Evo Engineering provides:

- system design
- process architecture
- deployment framework

Partners provide:

- capital
- operations
- logistics

Key Insight

Phoenix aligns disposal pressure with material demand, creating a favorable economic structure.

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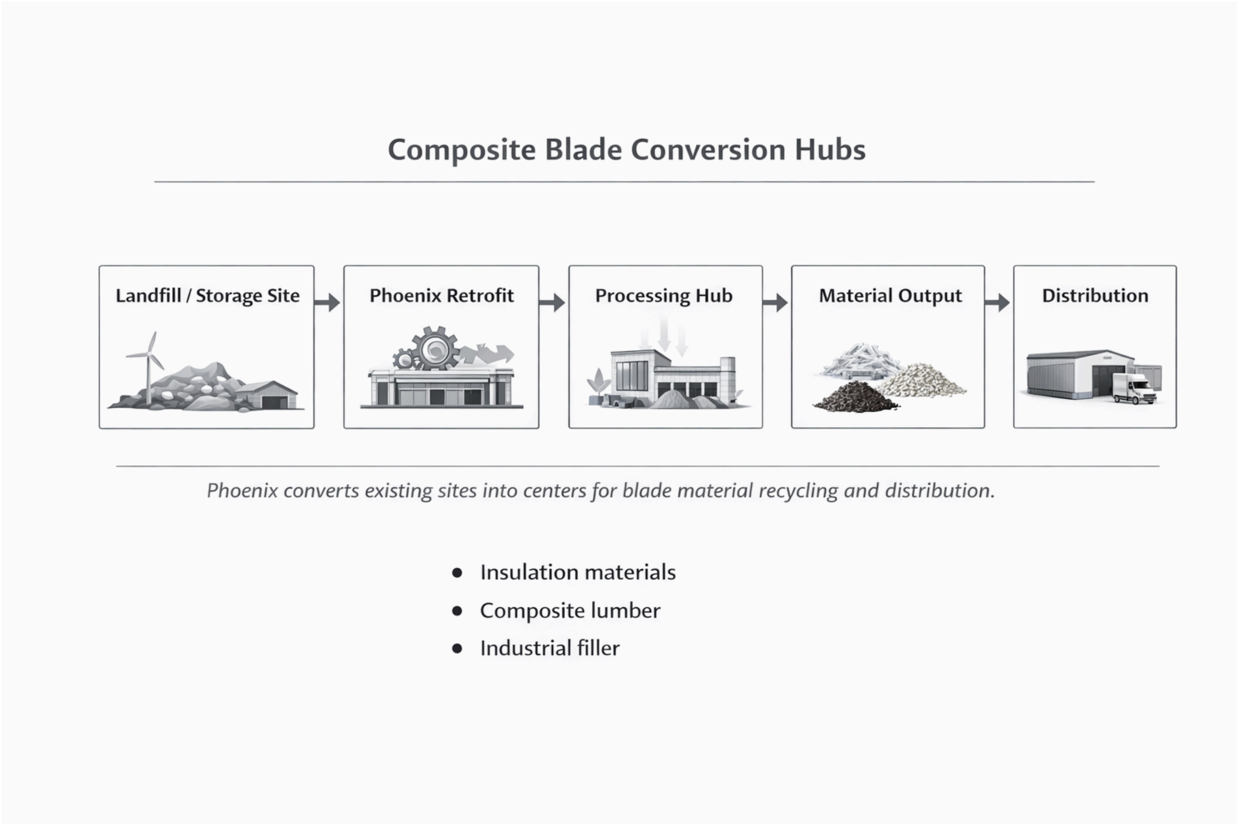
7. Deployment Model

Overview

Phoenix converts existing sites into:

Composite Blade Conversion Hubs



Advantages

- Low capital requirement
- Immediate feedstock access
- Scalable replication
- Reduced transport costs

Key Insight

Phoenix converts existing infrastructure into productive systems.

8. Pilot Plan

Objective

Validate system performance under real-world conditions.

Requirements

- Blade inventory
- Suitable site
- Operating partner

Outcomes

- Yield validation
- Cost per ton
- Throughput
- Market viability

Final Statement

Phoenix converts a growing disposal liability into a repeatable industrial revenue system